

Existing UI Surfaces + Design System — As-Is (multi-account research §12)

Revision: 1 **Last modified:** 2026-07-10T11:06:45Z **Status:** research-in-progress (as-is discovery — §11.4.74 catalogue-first, §11.4.6 evidence-only) **Authority:** operator mandate 2026-07-10 (multi-account BACKGROUND research); resolves 00_INDEX.md §5 **OQ-1 (OpenCode vs OpenDesign) Scope:** every user-facing UI surface (web dashboard + client apps) + the design system actually in use. READ-ONLY; no source edited, no git run.

0. TL;DR (what the evidence shows)

- **Two web/desktop UI surfaces ship today**, both React SPAs, both already with a **working sign-in + RBAC**: the **operator dashboard** (dashboard/, React 18 + Vite + react-router 6, hand-rolled components) and the **OTA Manager** (clients/ota-manager/, React 19 + Vite 6 + **shadcn/ui** + Tauri v2 desktop, also served as a web SPA at /manager).
- **The design system in actual use is the OpenDesign design-token package** design-systems/helix-ota/ (an od-design-system-project/v1 brand package), whose tokens.css is **vendored into both surfaces**. shadcn/ui sits *on top of* those tokens in OTA Manager; the dashboard consumes the same tokens through a hand-rolled ui.tsx. The OpenDesign **daemon/MCP** is a reviewed-but-not-yet-set-up recipe (operator-gated); only the **token files** are live.
- **“OpenCode” is NOT a UI or design system**. All 3299 in-repo hits live only under submodules/llm_orchestrator/ + submodules/vision_engine/, where it is an **LLM coding-agent adapter** (opencode_agent.go, sibling to junie/gemini/claudecode/qwencode). **OQ-1 resolved:** OpenCode = a coding *agent* used to *build* UI; OpenDesign = the mandated design-*token* system the built UI must consume (§11.4.162). Flagged for operator confirmation (§11.4.66) since the mandate said the literal word “OpenCode”.

- **No account/tenant concept exists in any UI** (0 account/tenant refs in either surface's src). But **OTA Manager already has a super_admin role, a per-project ProjectAccess{project_id, role} model, and a (mock) ProjectSwitcher** — the exact seam an account-switcher extends.
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1. Inventory of user-facing UI surfaces

Grounded by the prior static audit docs/research/opensdesign_ui_surface_audit_20260709/AUDIT.md (\$2 per-surface table) plus direct reads below.

Surface A — Operator Dashboard (dashboard/)

- **Identity:** dashboard/package.json:2 name ota-dashboard; :5 description “Helix OTA 1.0.0-MVP operator dashboard — React SPA over /api/v1”.
- **Framework:** React 18.3.1 + react-dom + **react-router-dom 6.28** (dashboard/package.json deps); **Vite 5** build; **no shadcn** — hand-rolled primitives in dashboard/src/components/ui.tsx; inline-style screens under dashboard/src/screens/ (LoginScreen, OverviewScreen, FleetScreen, ReleasesScreen, DeploymentsScreen, GroupsScreen, AuditScreen, ArtifactUploadScreen).
- **Platform:** web SPA served over /api/v1.
- **Routing/shell:** dashboard/src/App.tsx (AuthProvider → BrowserRouter → ProtectedRoute → AppShell); fixed nav list AppShell.tsx:36-44.

Surface B — OTA Manager (clients/ota-manager/)

- **Identity:** clients/ota-manager/package.json name ota-manager, description “Helix OTA Manager — Tauri v2 desktop client for managing OTA updates”.
- **Framework:** React 19 + **Vite 6** + **shadcn/ui** (components.json \$schema=shadcn, style:default, baseColor:slate, cssVariables:true) + **Tailwind v4**
 - **Radix** primitives (@radix-ui/*) + **TanStack** Router/Query/Table + **Zustand** + react-hook-form + zod. Components under src/components/ui/* (button, card, dialog, dropdown-menu, form, table, tabs, toast, ...). Feature folders under src/features/*.
- **Platform (dual): Tauri v2 desktop** (src-tauri/, tauri.conf.json productName “Helix OTA Manager”, macOS/Win/Linux) **AND** a

web SPA served by the Go server at /manager — server/internal/api/embed.go:56 //go:embed manager-dist/*; the SPA is mounted at /manager with a client-route fallback, API stays at /api/v1 (embed.go:18-27). So OTA Manager reaches users both as a desktop app and as the server-hosted /manager web console.

Surface C — Control-plane server (server/, Go/Gin)

- **No independent UI.** It only *serves* the OTA Manager build at /manager (embed.go); AUDIT §2 #3 confirms no Go HTML/template rendering. Not a separate design surface.

Surfaces D/E — Android bricks (no visual UI)

- submodules/ota-android-agent/ and submodules/ota-update-engine-bridge/ are **Kotlin, headless** — AUDIT §2 #4/#5: **0** Compose/@Composable/MaterialTheme markers, no res/**, no Activity/LAUNCHER (agent manifest declares only permissions + sharedUserId=android.uid.system). **There is no on-device UI today.** My own scan found Compose markers only in third-party vendored trees under submodules/helixqa/tools/opensource/**, never in the OTA Android bricks.

Out-of-scope surfaces (not Helix-OTA brand)

Per AUDIT §2 #6-11 + “Excluded”: HelixQA VitePress docs site (submodules/helixqa/website/), the **separate LLM-Verifier product** (submodules/llms_verifier/ — Angular web + Tauri/Electron desktop + RN/Flutter/Aurora/Harmony mobile, owned org vasic-digital, its own design system), plain Go CLIs (cmd/*, tools/*), and the Playwright e2e harness. None is a Helix-OTA account UI target.

Net: exactly two in-scope UI surfaces to extend — the dashboard (A) and OTA Manager (B); B doubles as the server’s /manager web console (C). Any device-side “setup wizard” (00_INDEX.md item 10) would be a *new* surface (D/E have no UI to extend).

2. Design system actually in use

2.1 The token package

design-systems/helix-ota/ is an **OpenDesign design-system brand package**: `manifest.json` `schemaVersion: "od-design-system-project/v1"`, `id: "helix-ota"`, `category: "Enterprise"`, `files: {design: DESIGN.md, tokens: tokens.css, tailwind: tailwind-v4.css}`. Its `tokens.css:1-31` provenance header states every brand color is **derived from the de-facto canonical Helix token set in clients/ota-manager/src/index.css** (shadcn/ui slate/blue theme, HSL → hex, losslessly), with **light and dark both first-class** (`:root = light`; dark via `@media (prefers-color-scheme: dark) + :root[data-theme="dark"] + .dark`).

2.2 How each surface consumes it

- **OTA Manager — ADOPTED.** `clients/ota-manager/src/index.css:2` `@import "../styles/opendesign-tokens.css"` (the OpenDesign semantic layer); that vendored file's header (`opendesign-tokens.css:2`) is `design-systems/helix-ota/tokens.css`. The shadcn HSL `:root` (`index.css:19+`, e.g. `--primary: 217.2 91.2% 59.8%`) intentionally wins 3 name collisions (documented `index.css:2` comment). AUDIT §2 #1: **0 hardcoded hex** in `src` outside the vendored token file. So shadcn/ui components render over the OpenDesign token layer.
- **Dashboard — PARTIAL.** `dashboard/src/main.tsx:6` `import "../styles/tokens.css"`; `dashboard/src/styles/tokens.css:1-2` header is **the same** `design-systems/helix-ota/tokens.css` (a vendored copy). `theme.ts` drives explicit data-theme light/dark. `ui.tsx` repoints tokens to `var(--...)`. But AUDIT §2 #2 records **58 hardcoded hex still remaining** in `src` (notably `AppShell.tsx:105/113/119/125` fixed brand chrome; the `AuditScreen/ReleasesScreen/ArtifactUploadScreen` tables) — the top remaining vendoring task. I confirmed the fixed-chrome hexes are deliberate (`AppShell.tsx:96-107` block comment: the dark-navy header is fixed in both themes).

2.3 Is an OpenDesign daemon already set up?

No — token files only. Per `docs/research/opendesign_integration_20260709/INTEGRATION_GROUND_TRUTH.md` §0: OpenDesign (`github.com/nexu-io/open-design`, tag `0.14.1`) is **NOT an npm CSS/component library** — it is a **local-first design product**

(pnpm monorepo, Electron app + od daemon + MCP server) that ships **token packages as plain files**. Adoption is a combination of (1) wiring the od daemon + MCP into the coding agent and (2) consuming the token files. Only **(2) is live** (the vendored `tokens.css`). Step (1) is a **reviewed-but-unbuilt, operator-gated recipe**: `docs/research/opendesign_daemon_setup_20260709/DAEMON_SETUP_RECIPE.md` status = “REVIEWED RECIPE, not an execution log — nothing here was built or run”.

Design system in use = the helix-ota OpenDesign token package (light+dark), consumed by both surfaces; shadcn/ui provides OTA Manager’s components; the dashboard uses a hand-rolled `ui.tsx` over the same tokens.

3. OQ-1 resolved — “OpenCode” vs “OpenDesign”

Evidence. `grep -rIi "opencode" (excl. node_modules/.git) = 3299 hits`, all confined to `submodules/llm_orchestrator/` and `submodules/vision_engine/`. The substantive code is an **LLM coding-agent adapter**: `submodules/llm_orchestrator/pkg/agent/opencode_agent.go` (+ `opencode_agent_unix.go`, `pkg/adapters/opencode_headless_test.go`), sitting **beside** `junie_agent.go`, `gemini_agent.go`, `claudecode_agent_test.go`, `qwencode_agent.go` — i.e. OpenCode is one of several interchangeable coding agents the orchestrator can drive. **Zero** “OpenCode” occurrences in `dashboard/`, `clients/`, `design-systems/`, or `docs/research/accounts/`.

Corroboration. The OpenDesign daemon’s job is to “**spawn coding-agent runtimes (Claude Code / Codex / opencode) as child processes** to do design work” (`DAEMON_SETUP_RECIPE.md` §0). So *within the tooling already in this repo*, **OpenCode is a coding agent that OpenDesign itself can invoke.**

Verdict (confirms the 00_INDEX.md §5 hypothesis): - **OpenCode = a coding agent** (a tool used to *author/build* UI code), analogous to Claude Code / Junie / Gemini-CLI. - **OpenDesign = the mandated design-token system** the built UI must consume (Constitution §11.4.162), already vendored as `design-systems/helix-ota/tokens.css`. - **Both can be true simultaneously** and are not in conflict: build the UI *with* OpenCode, style it *with* OpenDesign tokens. - **§11.4.66 flag for the operator:** the mandate’s literal phrasing “build the UI with OpenCode” should be confirmed to mean “the OpenCode coding agent” (not a distinct design

tool). Nothing named “OpenCode” is a UI/design library anywhere in this repo, so no guess is being substituted for that confirmation (§11.4.6).

4. Sign-in / auth UI today

Both in-scope surfaces already ship a working sign-in + session + RBAC (so this is *not* an API-token-only System at the UI layer):

Dashboard (A): - `dashboard/src/screens/LoginScreen.tsx` — OAuth2 ROPC form (username(email) + password → POST /auth/login). Single-factor; no account picker; no super-user vs user distinction. - `dashboard/src/auth/AuthContext.tsx` — full session: access JWT in-memory, refresh with rotation, single-in-flight refresh, route guarding, `hasRole()`. - `dashboard/src/types/api.ts:12` Role = "admin" | "operator" | "viewer" | "device" (note: **no super_admin here** — diverges from OTA Manager, see §6). - `dashboard/src/components/AppShell.tsx` — `RoleGate` (:29, UX-only), user area in the header (:72-87: subject + roles + logout + theme toggle). No account element.

OTA Manager (B): - `clients/ota-manager/src/features/auth/login-page.tsx` — email + password (react-hook-form + zod); `login-form.tsx`, `auth-guard.tsx` alongside. - `clients/ota-manager/src/stores/auth-store.ts` — Zustand persist store: token, refreshToken, `user{id,email,display_name,avatar_url,roles[],permissions[]}`, `isAuthenticated`. - `clients/ota-manager/src/features/layout/topbar.tsx` — user menu (:49-67: display_name, email, “Sign out”). No account element. - **RBAC already models super-user + per-project scope:** `clients/ota-manager/src/lib/api-client.ts:323` Role = 'viewer' | 'operator' | 'admin' | 'device' | 'super_admin'; :325-328 `ProjectAccess { project_id: string; role: Role }`; `clients/ota-manager/src/lib/permissions.ts` — `ROLE_HIERARCHY` with `super_admin:3` (:12-18), `isSuperAdmin()` (:143), `hasProjectAccess()/projectRole()` (:94-110), and a `projects` resource whose delete requires `super_admin` (:41).

Gap (auth UI). Neither surface has (a) a **super-user vs user** sign-in distinction (one login form each, one global identity space), nor (b) any **account selection after sign-in**, nor (c) a **super-admin administration console**. The `super_admin` role and per-project scope

exist in OTA Manager's *permission model* but have **no UI**. Confirmed by 0 account/tenant references in `clients/ota-manager/src` + `dashboard/src`.

5. Account-switcher precedent

OTA Manager already has a context switcher — `clients/ota-manager/src/features/layout/project-switcher.tsx`: a `shadcn` `DropdownMenu` “Projects” switcher (a `FolderKanban` trigger + checkmarked items). It is placed in the sidebar header (`features/layout/sidebar.tsx:15,57`, shown when the sidebar is expanded). **Today it is a stub** — `MOCK_PROJECTS` hardcoded (:13-16, *ATMOSphere / Helix OTA*) and local `useState` only. Its data hook already anticipates a server endpoint: `clients/ota-manager/src/hooks/use-projects.ts` queries `GET /projects` and falls back to a single placeholder “default” project until the server implements it (:21-45), keyed by `Project { project_id, name, description?, created_at, updated_at }` (`api-client.ts:568-574`).

This ProjectSwitcher + ProjectAccess + use-projects triad is the precedent an account-switcher extends — the account layer sits **above** project (`account` → `projects` → OTA updates, per *00_INDEX.md §4*). An `AccountSwitcher` would mirror `project-switcher.tsx` exactly (a second dropdown, above the project one, whose selection re-scopes which projects the project-switcher lists).

The dashboard (A) has no switcher of any kind — `AppShell.tsx:36-44` is a fixed route list and there is no project/account concept in the dashboard at all. A switcher there is greenfield.

6. Extension points for multi-account UI

Per surface, the concrete seam where **super-user** + **user sign-in** and **account selection** attach. Every new element **MUST** consume *OpenDesign* tokens (§11.4.162 — both surfaces already vendor `helix-ota/tokens.css`) and be proven by device-independent host-rendered pixels (§11.4.170 — harnesses already exist: `clients/ota-manager/visual/` and `dashboard/hostrender/`).

OTA Manager (clients/ota-manager/) — the richer, further-along surface: 1. **Account selection after sign-in** → add an AccountSwitcher paralleling features/layout/project-switcher.tsx, placed in features/layout/topbar.tsx (or the sidebar.tsx header above the project switcher); back it with a use-accounts.ts hook mirroring hooks/use-projects.ts. 2. **Super-user vs user** → sign-in stays features/auth/login-page.tsx; the distinction is post-auth: auth-store.ts already carries roles[]+permissions[] — extend the store with an **active account id**, and gate a super-admin console with permissions.ts:isSuperAdmin() (:143), which already exists. 3. **Super-admin console** → new TanStack route + a sidebar.tsx nav item gated by isSuperAdmin; the ProjectAccess{project_id, role} model (api-client.ts:325) generalises to AccountAccess{account_id, role} (or membership rows). 4. **Data model already project-scoped** → the switcher/hook/permission plumbing is present; the work is (a) real server endpoints for accounts + memberships, (b) wiring the switcher to them, (c) threading the active account into api-client requests. Because the server serves this build at /manager (embed.go:56), these changes reach the web console automatically on re-embed.

Dashboard (dashboard/) — greenfield for tenancy, needs alignment first: 1. **Sign-in seam** = src/screens/LoginScreen.tsx (already OAuth2 ROPC); **account picker** = a new post-login step + an AccountSwitcher in the AppShell.tsx header user area (:72-87) — there is no switcher precedent here to copy, so port the OTA-Manager pattern. 2. **Role model divergence** → types/api.ts:12 lacks super_admin (OTA Manager has it); align the two Role unions before adding super-admin UI. 3. **Finish token vendoring first** → repoint the remaining **58 hardcoded hex** (AUDIT §2 #2) to var(--token) before layering new tenancy UI, so new screens are OpenDesign-clean from the start.

Server /manager (C): no separate work — it is the OTA-Manager build; re-embed after OTA-Manager rebuilds.

Android/bridge (D/E): no UI to extend. If the device-side “setup wizard” (00_INDEX.md item 10) is built as an on-device Android surface, it is a *new* Compose UI — and AUDIT §2 #4 notes there is **no Compose theme at all** today, so it would need a **Kotlin Color.kt token bridge** off helix-ota/tokens.css + Roborazzi/Paparazzi host-render proof (§11.4.170). CSS tokens do not cross to Compose automatically.

Files read (provenance)

Repo root: /home/milos/Factory/projects/tools_and_research/
helix_ota. Every claim above cites one of these (paths absolute-from-root):

Dashboard (Surface A) - dashboard/package.json (name/description/deps: React 18, react-router-dom 6, Vite 5) - dashboard/src/App.tsx (router/shell composition, fixed route map) - dashboard/src/components/AppShell.tsx (nav :36-44, RoleGate :29, user area :72-87, fixed-chrome comment :96-107) - dashboard/src/screens/LoginScreen.tsx (OAuth2 ROPC login form) - dashboard/src/auth/AuthContext.tsx (session/JWT/refresh/guard) - dashboard/src/types/api.ts:12 (Role union — no super_admin) - dashboard/src/main.tsx:6 + dashboard/src/styles/tokens.css:1-2 (vendored helix-ota tokens)

OTA Manager (Surface B) - clients/ota-manager/package.json (React 19, Vite 6, shadcn deps, Tauri v2) - clients/ota-manager/components.json (shadcn config, baseColor slate, cssVariables) - clients/ota-manager/src/index.css:1-30 (@import opendesign-tokens.css :2; shadcn HSL :root) - clients/ota-manager/src/styles/opendesign-tokens.css:1-20 (header = helix-ota tokens.css) - clients/ota-manager/src/lib/api-client.ts:323-328,568-574 (Role+super_admin, ProjectAccess, Project) - clients/ota-manager/src/lib/permissions.ts (RBAC, super_admin, isSuperAdmin, project scoping) - clients/ota-manager/src/stores/auth-store.ts (Zustand session/user) - clients/ota-manager/src/features/auth/login-page.tsx (email/password sign-in) - clients/ota-manager/src/features/layout/{project-switcher,topbar,sidebar}.tsx (switcher precedent + user menu) - clients/ota-manager/src/hooks/use-projects.ts (GET / projects + placeholder) - clients/ota-manager/src/types/api.ts (re-exports incl. ProjectAccess, Project, Role)

Server / Android - server/internal/api/embed.go:12-56 (/go:embed manager-dist/*, /manager mount, /api/v1 separate) - submodules/ listing + Kotlin/Compose grep (Android bricks headless; Compose only in vendored helixqa/tools/opensource/**)

Design system + OpenCode/OpenDesign resolution - design-systems/helix-ota/manifest.json (od-design-system-project/v1, id helix-ota) - design-systems/helix-ota/tokens.css:1-209 (provenance, light+dark) - docs/research/opendesign_ui_surface_audit_20260709/AUDIT.md (§2 per-surface adoption table — grounds §1/§2) - docs/research/opendesign_integration_20260709/INTEGRATION_GROUND_TRUTH.md §0-§1 (OpenDesign = product+tokens, not npm lib) - docs/research/

opendesign_daemon_setup_20260709/DAEMON_SETUP_RECIPE.md §0-§1
(daemon reviewed-not-built; spawns “opencode” child agent) - docs/
research/frontend_production_readiness_20260709/READINESS.md head
(scope = ota-manager + dashboard + headless Android) - grep -rIi
opencode (3299 hits, all under submodules/llm_orchestrator +
submodules/vision_engine); submodules/llm_orchestrator/pkg/agent/
opencode_agent.go (+ _unix, pkg/adaptor/opencode_headless_test.go) -
docs/research/accounts/00_INDEX.md (mandate + OQ-1 statement this
doc resolves)

Honest gaps (§11.4.6)

- **Static inspection only.** This catalogues *what code exists*; it does **not** run any build or render any pixels. “Design system adopted” means “tokens are imported” (file:line), **not** that any surface renders correctly — that is the per-surface §11.4.170 host-render proof, out of scope here (prior proofs are referenced by AUDIT.md/READINESS.md, not re-verified by me).
- **OpenCode meaning is inferred from repo evidence, not an operator statement.** The determination (coding agent, not design tool) rests on where the token appears in the code + the OpenDesign daemon’s documented use of “opencode” as a child agent. Since the mandate used the literal word “OpenCode”, this is surfaced for operator confirmation per §11.4.66 — **UNCONFIRMED** until the operator confirms, but not a guess: no UI/design artifact named “OpenCode” exists anywhere in the tree.
- **Server-side account/tenant model not audited here.** I verified **0 account/tenant refs in the two UI src trees**; whether the Go server or store.Repository has any latent tenant seam is 10_existing_auth_and_project_model.md’s job, not this doc’s.
- **ProjectAccess/ProjectSwitcher are today stubs.** use-projects.ts returns a placeholder and project-switcher.tsx uses MOCK_PROJECTS; the per-project role model is real in the type layer but **not proven wired to a live server endpoint** — the server / projects endpoint’s existence is out of scope here (see 11_*/10_*).
- **Dashboard vs OTA-Manager role divergence is a real finding, not resolved here.** dashboard Role omits super_admin; reconciling the two role unions is design work for 21_authz_rbac_superuseradmin.md.